
PalmSource™ 99
Developing Automated Installations on Windows
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Overview

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- The APIs
  - HSAPI—The HotSync API
  - UserData—The users.dat API
  - CondMgr—The Registry API
  - InstAide—The Program Installation API
- Example (InstallShield)
- OCXs (ActiveX controls)
- General Issues

The HotSync API HSAPI.dll

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What You Can Do with It

- Stop, Start, Restart, or Refresh HotSync
  - Useful when adding conduits to HotSync
- Call HotSync Dialogs
  - Customize Conduits, Change Network Settings,
  - Display the HotSync Log, File Link, etc.
- Check/Set the Connection Status Type
  - Serial/Modem/Network connections
- Check the Sync Status
  - Idle/Syncing

The User Data API
UserData.dll

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Programmatic Access to Users.dat

- Add/Remove Users
- Set User Sync Preferences
- Modify User's Directory, Name, Password

- API Allows us to add functionality without breaking existing code

The Conduit Manager API CondMgr.dll

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The Big One

- Add/Remove Conduits
- Modify Conduits
  - Name, Directory, Priority, Title, etc.
- Add/Remove Notifiers
- Modify the Backup Conduit

- Access additional information about Conduits

The Installation Aide API InstAide.dll

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Programmatically Install Programs onto the Device

- Add/Remove files to the Download Directory
- Query supported file types
- Check what is waiting to be installed

Example Installation Code

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Get our Bearings

```
function GetPalmDir()
   NUMBER nType, nSize, nResult;
   STRING svTemp;
begin
   if(UseDLL(SUPPORTDIR ^ FILE_CONDMGR_DLL)
   = 0 ) then
      //Loaded successfully from the
      //Installer support directory
      nSize = ISHIELD_STR_SIZE;
      // Call to CondMgr to get path.
      nResult = CmGetCorePath(
      PILOT_PATH_HOTSYNC, svTemp, &nSize);
```

Example Installation Code

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Get our Bearings (Cont.)

```
        // Unload the dll
        UnUseDll(SUPPORTDIR ^ FILE_CONDMGR_DLL);

    // … (check results)

    ParsePath (svPalmDir, svTemp, PATH);

    return TRUE;
end;
```

Example Installation Code

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Pre-Flight Checklist

```
function CheckHSVersion()
   NUMBER nVerResult;
   STRING svHSVersion;
begin
   //Get the HotSync Version
   VerFindFileVersion(FILE_HOTSYNC_EXE,
   svPalmDir, svHSVersion);
   nVerResult = VerCompare (svHSVersion,
   STR_VERS_2X, VERSION);
   //Check result
   if (nVerResult = EQUALS || nVerResult =
   GREATER_THAN) then return TRUE
      else return FALSE;
   endif;
end
```

Example Installation Code

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Get Ready to Install Conduit

```
function AddConduit()
   NUMBER nSize, nErr;
   STRING svCondName;
begin
   if(UseDLL(svPalmDir ^ FILE_CONDMGR_DLL)=0)
   then
      //Check for previous existing
      nSize = ISHIELD_STR_SIZE;
      nErr = CmGetCreatorName( MY_CREATOR_ID,
      svCondName, &nSize);
      if (nErr != ERR_NO_CONDUIT ||
          nErr != ERR_INVALID_CREATOR_ID) then
         //It already exists
         //Remove it
```

Example Installation Code

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Get Ready to Install Conduit (Cont.)

```
        if (CmRemoveconduitByCreatorID(
        MY_CREATOR_ID) <= 0) then
           // Failed to Remove it, can't go on
           MessageBox("Failed to remove
           previous conduit");
           abort;
        endif;
     // should be ready to add now
     endif;
     nErr = InstallConduit();
     UnUseDll(svPalmDir ^ FILE_CONDMGR_DLL);

     // … check return values
end;
```

Example Installation Code

Install the Conduit

```
function InstallConduit()
   NUMBER nErr;
   STRING svConduitID;
begin
   svConduitID = MY_CREATOR_ID;
   // required
   nErr = CmInstallCreator(svConduitID,
   CONDUIT_APPLICATION); //… check return val
   nErr = CmSetCreatorName(svConduitID,
   MY_DLL_NAME); // … check return val
   nErr = CmSetCreatorDirectory(svConduitID,
   MY_DIRECTORY); // … check return val
   nErr = CmSetCreatorFile(szConduitID,
   MY_FILE); // … check return val
```

Example Installation Code

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Install the Conduit (Cont.)

```
    nErr = CmSetCreatorModule(svConduitID,
    MY_DLL); // … check return val
    nErr = CmSetCreatorRemote(svConduitID,
    MY_REMOTEDB); // … check return val
    // optional
    nErr = CmSetCreataorPriority(svConduitID,
    MY_PRIORITY); // … check return val

    nErr = CmSetCreatorIntegerate(svConduitID,
    MY_INTEGRATE); // … check return val

    return nErr;

end;
```

Example Installation Code

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Get the Username

```
function GetUserName(NUMBER nIndex)
   NUMBER nSize, nUserID, nErr;
   STRING svUserName;
begin
   if (UseDll(svPalmDir ^ FILE_USER_DATA)
   = 0) then
      nErr = UmGetUserID(nIndex, &nUserID);
      // … check return val
      nSize = ISHIELD_STR_SIZE;
      nErr = UmGetUserName(nUserID,
      svUserName, &nSize);
      UnUseDll(svPalmDir ^ FILE_USER_DATA);
   // … check return values
   return svUserName;
end;
```

Example Installation Code

Install the PRC

```
function InstallFile(STRING svUserName)
begin
   if (UseDll(svPalmDir ^ FILE_INST_AIDE)
   = 0) then
      nErr = PltInstallFile(svUserName,
      MY_PRC_FILE);
      // … check return val
      UnUseDll(svPalmDir ^ FILE_INST_AIDE);
   endif;
end;
```

Example Installation Code

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Stop HotSync

```
Function StopHS()
   NUMBER nErr;
begin
   if (UseDll(svPalmDir ^ FILE_HSAPI_DLL)
   = 0) then
      nErr = HsSetStatusApp(HS_CLOSE_APP,
      HSFLAG_NONE);
      UnUseDll(svPalmDir ^ FILE_HSAPI_DLL);
      if (nErr = 0) then
         return;
      endif
   endif
   MessageBox("Unable to stop HotSync");
   return;
end;
```

Example Installation Code

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Start HotSync

```
Function StartHS()
   NUMBER nErr;
begin
   if (UseDll(svPalmDir ^ FILE_HSAPI_DLL)
   = 0) then
      nErr = HsSetStatusApp(HS_START_APP,
      HSFLAG_NONE);
      UnUseDll(svPalmDir ^ FILE_HSAPI_DLL);
      if (nErr = 0) then
         return;
      endif
   endif
   MessageBox("Unable to start HotSync");
   return;
end;
```

Example Installation Code

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It All Comes Together

```
STRING svUserName;
program
   // … do all the setup and file moving
   GetPalmDir();                  // our CondMgr.dll
   StopHS();                      // HSAPI.dll
   if (CheckHSVerion() = FALSE) then
      abort;
   endif;
   AddConduit();                  // CondMgr.dll
   svUserName = GetUserName(USER_INDEX);
    // UserData.dll
   InstallFile(svUserName);// InstAide.dll
   StartHS();                     // HSAPI.dll
```

ActiveX Control PalmCntl.ocx

Wrapper for the APIs

- Provides all the APIs functionality to any COM compliant client (VB)

- Write your installers in other scripting languages beside InstallShield

General Issues

- Chicken/Egg issues of finding the Palm directory
- Don’t install directly into the Palm directory, a reinstall will not copy you
- Be Polite (Remove a previously existing conduit before installing a new one)
- Clean up after yourself (remove registered conduits when uninstalling)
- DON’T DIRECTLY ACCESS THE REGISTRY!!!


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